```
aws_wafv2_web_acl_association.app_waf_assoc: Still creating... [10s elapsed]
aws_wafv2_web_acl_association.app_waf_assoc: Still creating... [20s elapsed]
aws_wafv2_web_acl_association.app_waf_assoc: Creation complete after 30s [id=arn:aws:wafv2:us-west-2:129762072419:regional/webacl/lab1-redone-waf/c19b09df-7396-4a7d-b1ff
-1b26ee13f94c,arn:aws:elasticloadbalancing:us-west-2:129762072419:loadbalancer/app/lab1-redone-alb/2d2507feff869820]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

alb_dns_name = "lab1-redone-alb-879803269.us-west-2.elb.amazonaws.com"
app_instance_id = "i-01772e88ccd210f98"
app_instance_profile_name = "lab1-redone-app-instance-profile"
app_role_name = "lab1-redone-app-role"
app_url = "http://tetsuzai-kube.com"
db_secret_name = "lab1-redone/rds/mysql"
private_db_subnet_ids = [
  "subnet-0d353663c0592be25",
  "subnet-050c888c309ac728f",
]
public_subnet_ids = [
  "subnet-0e5f3be95f460ba57",
  "subnet-06fe10f961ad39a2e",
]
rds_endpoint = "lab1-redone-mysql.cn8sqw08svwd.us-west-2.rds.amazonaws.com"
vpc_id = "vpc-0996cfa3e351ffae2"
```

I implemented AWS WAF in front of the application load balancer using AWS managed rule groups to provide baseline web-layer protection against common attack patterns.

```
Outputs:

alb_dns_name = "lab1-redone-alb-879803269.us-west-2.elb.amazonaws.com"
app_instance_id = "i-01772e88ccd210f98"
app_instance_profile_name = "lab1-redone-app-instance-profile"
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app_instance_profile_name = "lab1-redone-app-instance-profile"
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]
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  "subnet-0e5f3be95f460ba57",
  "subnet-06fe10f961ad39a2e",
]
rds_endpoint = "lab1-redone-mysql.cn8sqw08svwd.us-west-2.rds.amazonaws.com"
vpc_id = "vpc-0996cfa3e351ffae2"
(base) PS C:\Users\marvi\Desktop\armageddon_labs_2026\LAB1\lab1a\terraform>
```

WAF & Shield

AWS WAF
Protection packs (web ACLs) New
IP sets
Regex pattern sets
Rule groups
Application integration
Add-on protections

AWS Shield
Getting started
Overview
Protected resources
Events
Global threat dashboard

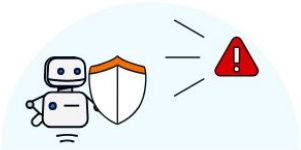
AWS Shield network security director
Getting started Preview
Dashboard

AWS Firewall Mana
Getting started
Security policies
Resource sets
Application list
Protocol lists
Settings

Resources and protection packs (web ACLs)

View your secured assets and their active security controls.

How it works

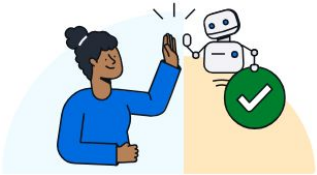


Protect your resources with packs

Protection packs are the new location for your web ACLs. Here you can supercharge your protections.

Watch live traffic

Use the graphs and dashboard to see how your resources are being affected. See your protections safeguard your resources in real time.



Smart recommendations

Based on traffic potential vulnerability, and aggregate data, we will give you tailor made recommendations to stop malicious actors before they attack.

Protection packs (web ACLs) (7)

Actions | Create protection pack (web ACL)

Use protection packs (web ACLs) to combine rules and rule sets that protect your infrastructure from threats.

Search protection packs (web ACLs) by name, id, scope, etc.

Region scope: CloudFront (Global) and Regional

1 | List | Grid

Name	Rules	Resources	Dashboard	AI traffic analysis new	Sampled requ...	Logging & metrics	Scope	ARN
lab1-redone-waf	1 rules	Manage	View	View	View	Not enabled	Regional	arn:aws:wafv2:us-
tetsuzai-waf01	0 rules	Manage	View	View	View	Not enabled	Regional	arn:aws:wafv2:us-
tetsuzai-cloudfront-waf	0 rules	Manage	View	View	View	Not enabled	CloudFront (Global)	arn:aws:wafv2:us-

WAF & Shield > Protection packs (web ACLs) > lab1-redone-waf > Dashboard

lab1-redone-waf

Edit protection pack (web ACL)

Add recommended rules based on your traffic insights (4) Info

AWS Threat Intelligence monitors your previous 2 weeks of allowed traffic for vulnerabilities. Mitigate these by interacting with the specific traffic dimensions below to enable the recommended AWS Managed Rules.

Vulnerable application traffic	Allowed Bots traffic	Allowed DDoS sources	Allowed IP reputation traffic	Allowed Anonymous IP traffic
0	0	0	0	0

Filter data

Last 3 hours | Local time

Actions = ALLOW, BLOCK, CAPTCHA, CHALLENGE | Traffic type = ALL | Clear filters

Summary Info

Request counts for all traffic during the specified time range.

Total	Allowed	Blocked	CAPTCHA	Challenged

Sampled requests

All metrics | Search requests

Log explorer | Sampled requests

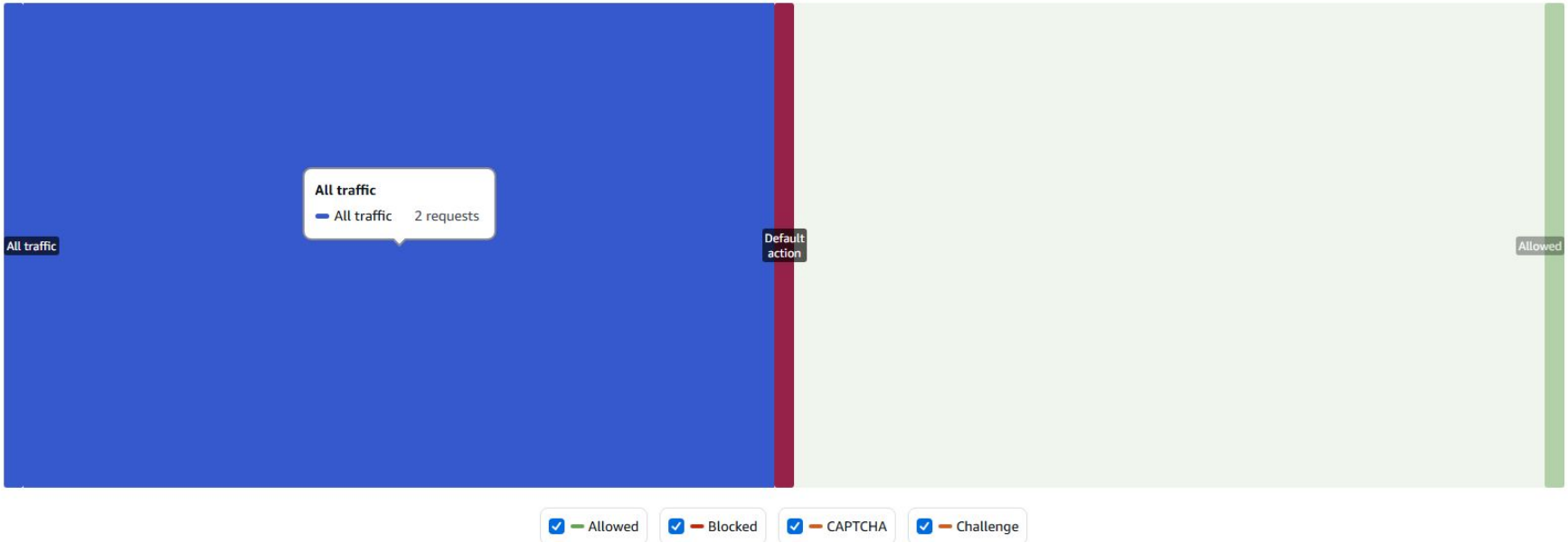
Metric name	Source IP	URI	Rule inside rule group	Action	Time
lab1-redone-waf	172.5.19.166	/	-	ALLOW	Mon Apr 13 2026 18:05:46 GMT-050...
lab1-redone-waf	172.5.19.166	/favicon.ico	-	ALLOW	Mon Apr 13 2026 18:05:46 GMT-050...

Actions = ALLOW, BLOCK, CAPTCHA, CHALLENGE | Traffic type = ALL | Clear filters

Protection pack (web ACL) activity Info

Sequential rules view | Non-sequential rules view

Summary of your protection rules and how their order contributes to terminating actions.



EC2 > Load balancers

AWS Global View

Events

Instances

Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New

Images

AMIs
AMI Catalog

Elastic Block Store

Volumes
Snapshots
Lifecycle Manager

Network & Security

Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing

Load Balancers
Target Groups
Trust Stores

Auto Scaling

Auto Scaling Groups

Load balancers (1/1) What's new?

Actions | Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

1

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS n
lab1-redone-alb	Active	application	Internet-facing	IPv4	vpc-0996cfa3e351ffae2	2 Availability Zones	sg-06a275a631b9fdde5	lal

Load balancer: lab1-redone-alb

Details | Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

You can integrate the following AWS services with your load balancer. Integration status and details are displayed below. You set up and configure integration through these services.

Amazon Application Recovery Controller (ARC) Info

Optimizes: **Availability**

Not integrated

Amazon CloudFront + AWS Web Application Firewall (WAF) Info

Optimizes: **Performance, Security, Availability**

Not integrated

AWS Global Accelerator Info

Optimizes: **Performance, Availability**

Not integrated

AWS Config Info

Optimizes: **Monitoring, Compliance**

Not integrated

AWS Web Application Firewall (WAF) Info

Optimizes: **Security**

Integrated

Integration status details
This load balancer is integrated with AWS WAF

AWS WAF web ACL
lab1-redone-waf

Resource-level DDoS protection
Active under DDoS

dashboard

```
Outputs:

alb_dns_name = "lab1-redone-alb-879803269.us-west-2.elb.amazonaws.com"
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app_instance_profile_name = "lab1-redone-app-instance-profile"
app_role_name = "lab1-redone-app-role"
app_url = "http://tetsuzai-kube.com"
db_secret_name = "lab1-redone/rds/mysql"
private_db_subnet_ids = [
  "subnet-0d353663c0592be25",
  "subnet-050c888c309ac728f",
]
public_subnet_ids = [
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  "subnet-06fe10f961ad39a2e",
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rds_endpoint = "lab1-redone-mysql.cn8sqw08svwd.us-west-2.rds.amazonaws.com"
vpc_id = "vpc-0996cfa3e351ffae2"
(base) PS C:\Users\marvi\Desktop\armageddon_labs_2026\LAB1\lab1a\terraform>
```

CloudWatch > Dashboards

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms 0 1 1

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Custom Dashboards | Automatic dashboards

Custom Dashboards (2) Info

Share dashboard | Delete | Create dashboard

Filter dashboards

1

Name	Sharing	Favorite	Last update (UTC)
lab1-redone-dashboard		☆	2026-04-13 23:18
tetsuzai-alb-dashboard		☆	2026-02-08 21:43

CloudWatch > Dashboards > lab1-redone-dashboard

Autosave: Off

lab1-redone-dashboard

1h 3h 12h 1d 3d 1w Custom | UTC timezone | Actions | Save

ALB Request Count

Count

3
1.5
0

20:30 20:45 21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15

RequestCount

ALB Target Response Time

Seconds

2 ms
1.7 ms
1.4 ms

20:30 20:45 21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15

TargetResponseTime

ALB Healthy / Unhealthy Hosts

Count

1
0.5
0

20:30 20:45 21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15

HealthyHostCount UnHealthyHostCount

ALB 5XX Errors

1
0.5
0

No data available.
Try adjusting the dashboard time range.

20:30 20:45 21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15

HTTPCode_ELB_5XX_Count

WAF Allowed Requests

3
2.5
2

20:30 20:45 21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15

AllowedRequests

WAF Blocked Requests

1
0.5
0

No data available.
Try adjusting the dashboard time range.

20:30 20:45 21:00 21:15 21:30 21:45 22:00 22:15 22:30 22:45 23:00 23:15

BlockedRequests

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Python verification

```
TERMINAL
(base) PS C:\Users\marvi\Desktop\armageddon_labs_2026\LAB1\lab1c\validation> python verify_lab1c.py
PASS: ALB exists
      lab1-redone-alb
PASS: WAF Web ACL exists
      lab1-redone-waf
PASS: WAF is associated with ALB
      lab1-redone-waf -> lab1-redone-alb
PASS: CloudWatch dashboard exists
      lab1-redone-dashboard
PASS: CloudWatch dashboard has widgets
      Widget count: 6

Validation report written to lab1c_validation.json
(base) PS C:\Users\marvi\Desktop\armageddon_labs_2026\LAB1\lab1c\validation>
```

TERMINAL

```
(base) PS C:\Users\marvi\Desktop\armageddon_labs_2026\LAB1\lab1c\validation> python verify_lab1c.py
PASS: WAF Web ACL exists
      lab1-redone-waf
PASS: WAF is associated with ALB
      lab1-redone-waf -> lab1-redone-alb
PASS: CloudWatch dashboard exists
      lab1-redone-dashboard
PASS: CloudWatch dashboard has widgets
      Widget count: 6

Validation report written to lab1c_validation.json
(base) PS C:\Users\marvi\Desktop\armageddon_labs_2026\LAB1\lab1c\validation> python verify_lab1c.py | tee validation_terminal_output.txt
PASS: ALB exists
      lab1-redone-alb
PASS: WAF Web ACL exists
      lab1-redone-waf
PASS: WAF is associated with ALB
      lab1-redone-waf -> lab1-redone-alb
PASS: CloudWatch dashboard exists
      lab1-redone-dashboard
PASS: CloudWatch dashboard has widgets
      Widget count: 6

Validation report written to lab1c_validation.json
(base) PS C:\Users\marvi\Desktop\armageddon_labs_2026\LAB1\lab1c\validation>
```